

ARYABHATTA KNOWLEDGE UNIVERSITY, PATNA

BACHELOR OF COMPUTER APPLICATIONS (BCA)

SEMESTER III

INTERNET & WEB DESIGNING

Paper Code: BCA-302 / 303302

Full Marks: 60 | Time: 3 Hours

PYQ Analysis: 2014 – 2025

Exam Study Notes — Unit-wise with Definitions & Key Points

CONTENTS

Unit	Topic	MCQs	4-Mark	12-Mark
1	Internet Basics / TCP-IP / Domain	10	5	2
2	Introduction to HTML	12	5	2
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★ **VVI — MOST REPEATED IN EXAM:** TCP/IP Layers (7x) • JSP Actions (6x) • JavaScript Adv/Disadv (6x) • HTML Structure (5x) • Page Directive (5x)

SECTION A — MCQ (Multiple Choice Questions)**Q1. Who is known as father of WWW?****[PYQ: 2014]**

- A. John T. Thomson
- B. Dennis Ritchie
- ✓ C. Tim Berners-Lee
- D. None of them

Explanation: Tim Berners-Lee invented the World Wide Web in 1989 at CERN, Switzerland.

Q2. Primary protocol used to distribute info on the web is:**[PYQ: 2014]**

- A. TCP/IP
- B. FTP
- ✓ C. HTTP
- D. Telnet

Explanation: HTTP (HyperText Transfer Protocol) is the main protocol of the World Wide Web.

Q3. Internet is defined as:**[PYQ: 2020]**

- A. A single network
- ✓ B. A vast collection of different networks
- C. Interconnection of LANs only
- D. None

Explanation: Internet = global network of millions of interconnected networks using TCP/IP.

Q4. To join the Internet, a computer connects to an:**[PYQ: 2020]**

- A. Internet Architecture Board
- B. Internet Society
- ✓ C. Internet Service Provider (ISP)
- D. None

Explanation: ISP provides the gateway connection to the Internet for home/office users.

Q5. IPv6 addresses have a size of:**[PYQ: 2017, 2019]**

- A. 32 bits
- B. 64 bits
- ✓ C. 128 bits
- D. 265 bits

Explanation: IPv6 = 128-bit addressing. IPv4 = 32-bit. IPv6 supports far more addresses.

Q6. Most widely used Internet Protocol version is:

[PYQ: 2018]

- A. IPv1
- B. IPv6
- C. IPv2
- ✓ D. IPv4

Explanation: IPv4 (32-bit) is still the most widely deployed IP protocol globally.

Q7. Internet uses which type of switching technology?

[PYQ: 2019]

- ✓ A. Packet switching
- B. Circuit switching
- C. Telephone switching
- D. Telex switching

Explanation: Internet breaks data into packets; each packet travels independently (packet switching).

Q8. Web pages are uniquely defined using:

[PYQ: 2019]

- A. IP addresses
- ✓ B. URL
- C. Domain
- D. File name

Explanation: URL (Uniform Resource Locator) is the unique address of each web page.

Q9. WWW is based on which model?

[PYQ: 2019]

- A. Local-server
- ✓ B. Client-server
- C. 3-tier
- D. None

Explanation: WWW follows Client-Server model — browser (client) requests, server responds.

Q10. Domain names are converted to:

[PYQ: 2018]

- A. Binary string
- B. Alphanumeric string
- ✓ C. IP addresses
- D. Hexadecimal string

Explanation: DNS (Domain Name System) translates domain names (e.g. google.com) to IP addresses.

SECTION B — SHORT ANSWER QUESTIONS (4 Marks)

Q1. What is Internet? Define with its main features.

[PYQ: 2020, 2022]

The Internet is a global network of millions of interconnected computers communicating using the TCP/IP protocol suite.

Main Features of Internet:

- **Global Reach:** Connects billions of devices and users worldwide
- **Packet Switching:** Data is broken into small packets for transmission
- **Open Standards:** Uses TCP/IP protocol, open to all networks
- **Services:** Email, WWW, FTP, Telnet, Video Conferencing, Chat
- **Decentralized:** No single authority controls the entire Internet
- **24x7 Access:** Available from anywhere, anytime in the world

Q2. What is TCP/IP? Explain its layers and main services.

[PYQ: 2022]

TCP/IP stands for Transmission Control Protocol / Internet Protocol. It is the foundational protocol suite of the Internet.

TCP — Transmission Control Protocol:

- Ensures reliable and ordered delivery of data packets
- Breaks data into segments and reassembles at destination
- **Uses 3-Way Handshake:** SYN to SYN-ACK to ACK

IP — Internet Protocol:

- Handles addressing and routing of packets across networks
- IPv4 = 32-bit addresses; IPv6 = 128-bit addresses
- Best-effort delivery (TCP adds the reliability on top)

Q3. What is a Web Server? Give examples.

[PYQ: 2022]

A Web Server is a computer/software that stores web content (HTML files, images, CSS) and delivers it to browsers using the HTTP/HTTPS protocol.

Key Functions:

- Accepts HTTP requests from browsers (clients)
- Processes the request and sends back HTML responses
- Runs 24x7 to serve web pages on demand

Examples of Web Servers:

- **Apache HTTP Server:** most popular, open-source, free
- **Nginx:** fast and lightweight web server
- **Microsoft IIS:** for Windows-based servers
- **Apache Tomcat:** for JSP and Java-based web applications

Q4. What is Domain? Explain Domain Registration process.

[PYQ: 2022]

A Domain is a human-readable name that maps to an IP address. For example: www.google.com. DNS (Domain Name System) translates it to an IP address.

Types of Domain Extensions:

- **.com:** Commercial websites; .edu: Educational institutions
- **.gov:** Government websites; .in: India (country code TLD)

Domain Registration Process:

- **Step 1:** Choose a domain name (unique, relevant)
- **Step 2:** Check availability through a Domain Registrar
- **Step 3:** Register through Registrar (GoDaddy, Namecheap, BigRock)
- **Step 4:** Pay yearly fee to own/renew the domain
- **Step 5:** Configure DNS records to point to your server IP

Q5. What is Private IP Address? What is Loopback address?

[PYQ: 2020, 2022]

A Private IP Address is an IP used within a private local network (LAN). It is not directly accessible from the public Internet.

Private IP Address Ranges:

- **Class A:** 10.0.0.0 to 10.255.255.255
- **Class B:** 172.16.0.0 to 172.31.255.255
- **Class C:** 192.168.0.0 to 192.168.255.255 (most commonly used at home)

Loopback Address:

- The loopback address is 127.0.0.1
- It refers to the local machine itself
- Used for testing network programs without a real network connection
- Typing 127.0.0.1 in a browser connects to your own computer

SECTION C — LONG ANSWER QUESTIONS (12 Marks)

Q1. Explain TCP/IP protocol in detail — all layers and services. (★ MOST REPEATED — 7 times)

[PYQ: 2017, 2018, 2019, 2020, 2022, 2024, 2025]

Introduction:

TCP/IP (Transmission Control Protocol / Internet Protocol) is the backbone protocol suite of the Internet. It defines how data is transmitted, addressed, routed, and received across interconnected networks.

TCP/IP has 4 Layers (Top to Bottom):

- **Layer 4 — Application Layer:** Provides services directly to applications. Protocols include: HTTP (web browsing), FTP (file transfer), SMTP (sending email), DNS (domain name lookup), Telnet (remote login)
- **Layer 3 — Transport Layer:** Manages end-to-end communication between programs. TCP provides reliable, ordered delivery. UDP provides fast but unreliable delivery. Uses port numbers to identify applications.
- **Layer 2 — Internet / Network Layer:** Handles logical addressing and packet routing. Key protocols: IP (addressing), ICMP (error messages like ping), ARP (converts IP to MAC address)
- **Layer 1 — Network Access Layer:** Handles physical transmission of data. Protocols: Ethernet (wired), Wi-Fi (wireless), PPP (dial-up)

Features of TCP:

- **Connection-oriented:** establishes connection before sending data
- **Reliable delivery:** uses acknowledgements (ACK) and retransmission
- **Flow control:** prevents sender from overwhelming receiver
- **Sequencing:** data packets reassembled in correct order at destination
- **3-Way Handshake:** SYN then SYN-ACK then ACK

TCP/IP Services:

- **WWW:** Web browsing through HTTP and HTTPS protocols
- **Email:** SMTP to send email; POP3 or IMAP to receive email
- **FTP:** File Transfer Protocol for uploading and downloading files
- **Telnet:** Remote login to another computer over the network
- **DNS:** Domain Name System converts domain names to IP addresses
- **DHCP:** Automatically assigns IP addresses to devices on a network

Conclusion:

TCP/IP is the most important protocol suite. Every device on the Internet uses TCP/IP. It makes global communication possible by providing reliable, layered, universal communication standards.

Introduction:

Domain registration and electronic mail (email) are two essential services of the Internet used daily by billions of people worldwide.

Domain Name System (DNS):

- DNS is like a phonebook for the Internet — converts names to numbers
- Every website has an IP address (e.g. 142.250.180.46 for Google)
- DNS allows us to type www.google.com instead of a number

Domain Registration — Step by Step:

- **Step 1:** Choose a domain name (must be unique, memorable)
- **Step 2:** Check availability using a Domain Registrar website
- **Step 3:** Register through registrars: GoDaddy, Namecheap, BigRock
- **Step 4:** Pay a yearly registration fee to own the domain
- **Step 5:** Configure DNS records — A record (IP), MX record (mail), CNAME
- **Step 6:** ICANN manages all top-level domains globally

Domain Extensions:

- **Generic TLD:** .com (commercial), .org (organization), .edu (education), .gov (government)
- **Country Code TLD:** .in (India), .uk (United Kingdom), .us (United States)

Working of Electronic Mail — Step by Step:

- **Step 1 — Compose:** User writes email in a Mail User Agent (Gmail, Outlook, Thunderbird)
- **Step 2 — Send (SMTP):** Simple Mail Transfer Protocol sends email from client to sender mail server
- **Step 3 — DNS MX Lookup:** Sender mail server queries DNS for recipient mail server using MX record
- **Step 4 — Transfer:** Email travels from sender server to recipient server through Internet
- **Step 5 — Receive (POP3/IMAP):** POP3 downloads email to device; IMAP keeps email on server
- **Step 6 — Read:** Recipient opens email in their Mail User Agent

Conclusion:

Domain registration and email are fundamental Internet services. DNS enables human-readable addresses while email enables instant global communication using SMTP, POP3, and IMAP protocols.

SECTION A — MCQ (Multiple Choice Questions)**Q1. Which tag is used to display an image on a web page?****[PYQ: 2014, 2024, 2025]**

- A. <image>
- ✓ B.
- C. <src>
- D. <pic>

Explanation: is the correct HTML tag. The tag name is img, not image.

Q2. Which tag makes text bold with semantic importance?**[PYQ: 2017]**

- A. <bold>
- ✓ B.
- C. <dar>
- D. <emp>

Explanation: makes text bold AND indicates semantic importance. makes text bold only.

Q3. Tags not displayed on page are written inside which section?**[PYQ: 2017]**

- A. <html>
- ✓ B. <head>
- C. <title>
- D. <body>

Explanation: The <head> section contains metadata, CSS links, scripts — nothing in head is visible.

Q4. Which tag inserts a horizontal line across the page?**[PYQ: 2017]**

- ✓ A. <hr>
- B. <line>
- C. <hl>
- D. <tr>

Explanation: <hr> = Horizontal Rule. It inserts a horizontal dividing line across the page.

Q5. First tag in any HTML document must be:**[PYQ: 2017]**

- A. <head>
- B. <title>
- ✓ C. <html>
- D. <body>

Explanation: <html> is the root element. It wraps the entire HTML document structure.

Q6. Which tag adds a new row in an HTML table?

[PYQ: 2017]

- A. <td>
- B. <row>
- C. <th>
- ✓ D. <tr>

Explanation: <tr> = Table Row. It contains <td> (data cells) or <th> (header cells).

Q7. How to create an unordered (bulleted) list in HTML?

[PYQ: 2017, 2024]

- A. <list>
- B. <nl>
- ✓ C.
- D.

Explanation: = Unordered List. Each item uses tag. creates numbered list.

Q8. Main function of a web browser is:

[PYQ: 2018]

- A. Compile HTML
- ✓ B. Interpret and render HTML
- C. Decompile HTML
- D. Interpret CGI

Explanation: Browser interprets HTML code and renders it as a visual web page on screen.

Q9. CSS property used to change space between text lines:

[PYQ: 2024, 2025]

- A. text-spacing
- ✓ B. line-height
- C. letter-spacing
- D. word-spacing

Explanation: line-height controls the vertical space between lines of text in CSS.

Q10. URL stands for:

[PYQ: 2024, 2025]

- A. Universal Resource Locator
- ✓ B. Uniform Resource Locator
- C. Unified Resource Link
- D. User Related Location

Explanation: URL = Uniform Resource Locator. It is the unique address of any web resource.

Q11. ALIGN in is:

[PYQ: 2018]

- A. A tag
- B. A head element
- C. A body element
- ✓ D. An attribute

Explanation: ALIGN is an attribute of the IMG tag. Attributes provide extra information about tags.

Q12. Which tag creates table header cells (bold and centered by default)?

[PYQ: 2018]

- A. <head>
- B. <tbody>
- ✓ C. <th>
- D. <td>

Explanation: <th> = Table Header cell. These appear bold and centered automatically.

SECTION B — SHORT ANSWER QUESTIONS (4 Marks)

Q1. Explain any four text formatting tags in HTML with examples.

[PYQ: 2018, 2019]

HTML text formatting tags change how text appears on the web page. Common formatting tags are:

- ****: Makes text Bold. Example — Bold Text
- ****: Bold text with semantic emphasis (important text)
- **<i>**: Makes text Italic. Example — <i>Italic Text</i>
- **<u>**: Underlines text. Example — <u>Underlined</u>
- **<sup>**: Superscript — raised text like x² in math
- **<sub>**: Subscript — lowered text like H₂O in chemistry
- ****: Changes font size, color, and face
- **<h1> to <h6>**: Heading tags — h1 is largest, h6 is smallest

Q2. Describe all types of lists used in HTML with examples.

[PYQ: 2022]

HTML supports three types of lists to organize information:

1. Unordered List — (Bulleted List):

- Creates bullet points before each item
- **type attribute:** disc (default), circle, square

```
<ul type="disc">
<li>Apple</li>
<li>Mango</li>
</ul>
```

2. Ordered List — (Numbered List):

- Numbers items automatically
- **type:** 1 (numbers), A (capitals), a (small), I (Roman), i (roman)

3. Definition List — <dl>

- **<dt>**: Definition Term (the word)
- **<dd>**: Definition Description (the meaning)

Q3. Explain how tables are created in HTML. Describe all table tags and attributes.

[PYQ: 2018, 2019, 2022]

HTML tables display data organized in rows and columns. Tables are created using a set of related tags:

Main Table Tags:

- **<table>**: Defines the table container
- **<tr>**: Table Row — defines one row inside the table
- **<th>**: Table Header — bold, centered header cell
- **<td>**: Table Data — regular data cell

Table Attributes:

- **border**: sets the border width (border="1")
- **cellpadding**: space inside each cell
- **cellspacing**: space between cells
- **colspan**: merge multiple columns into one cell
- **rowspan**: merge multiple rows into one cell
- **align**: alignment of table (left, center, right)

Q4. How are frames created in HTML? What is frameset? Write example.

[PYQ: 2020, 2014]

Frames divide the browser window into multiple sections. Each section shows a different HTML page. Frameset replaces the body tag.

Key Frame Tags:

- **<frameset>**: Replaces <body>; defines how frames are arranged
- **<frame src="page.html">**: Each frame with its own page
- **cols attribute**: splits window vertically (columns)
- **rows attribute**: splits window horizontally (rows)

```
<html>
<frameset cols="50%, 50%">
<frame src="left.html" name="leftframe">
<frame src="right.html" name="rightframe">
</frameset>
</html>
```

Q5. How to create links in HTML? Explain all types of hyperlinks with examples.

[PYQ: 2022]

Links (Hyperlinks) connect web pages together. The anchor tag <a> with href attribute creates links.

Types of Links:

- **External Link:** Links to another website — Google
- **Internal Link:** Links to another page in same site — About
- **Anchor Link:** Links to a section on the same page — Go to Section 1
- **New Tab Link:** Opens in new tab — target=" blank" attribute
- **Email Link:** Opens email program —

SECTION C — LONG ANSWER QUESTIONS (12 Marks)

Q1. What is HTML? Explain the complete structure of an HTML document with example. (★ 5 times)

[PYQ: 2020, 2024, 2025]

Introduction:

HTML stands for HyperText Markup Language. It is the standard language used to create and structure web pages. HTML uses tags to define the structure of web content displayed in a browser.

Key Features of HTML:

- Markup language — not a programming language
- Uses tags (commands in angle brackets) to structure content
- **Tags come in pairs:** opening tag <tag> and closing tag </tag>
- Not case-sensitive (but lowercase is recommended standard)
- Supports text, images, links, tables, forms, audio, video

Structure of an HTML Document:

```
<!DOCTYPE html>

<html>

<head>

<title>My Web Page</title>

<meta charset="UTF-8">

</head>

<body bgcolor="white">

<h1>Welcome to AKU Patna</h1>

<p>This is a paragraph of text.</p>

<a href="page2.html">Go to Page 2</a>

</body>

</html>
```

Each Part Explained:

- **<!DOCTYPE html>:** Declaration — tells browser this is an HTML5 document
- **<html>:** Root element — wraps the entire HTML document
- **<head>:** Contains metadata, title, CSS links (NOT visible on page)
- **<title>:** Text shown in browser tab or title bar
- **<meta charset>:** Specifies character encoding (UTF-8 supports all languages)
- **<body>:** Contains all visible content of the web page
- **<h1> to <h6>:** Heading tags from largest to smallest
- **<p>:** Paragraph tag for regular text blocks
- **
:** Line break (single tag, no closing needed)
- **<hr>:** Horizontal rule line across the page

Roles of HTML, CSS, and JavaScript:

- **HTML:** Structure / Skeleton of the web page
- **CSS:** Style / Appearance (colors, fonts, layout, spacing)
- **JavaScript:** Behavior / Interactivity (dynamic content, animations)

Conclusion:

HTML is the foundation of every web page. It is simple to learn and forms the base of all web development. Every website you visit is built using HTML as its structure.

Q2. What is an HTML Form? Explain all form elements with a complete registration form example.

[PYQ: 2018]

Introduction:

An HTML Form is used to collect input from users. The form tag wraps all form elements. Data is sent to a server for processing using the action (URL) and method (GET or POST) attributes.

Main Form Elements:

- `<input type="text">`: Single-line text input field
- `<input type="password">`: Password field (characters are hidden/masked)
- `<input type="radio">`: Radio button — select only ONE option from a group
- `<input type="checkbox">`: Checkbox — select ONE or MORE options
- `<input type="submit">`: Submit button to send form data
- `<input type="reset">`: Reset button to clear all form inputs
- `<textarea>`: Multi-line text input box
- `<select>` + `<option>`: Dropdown menu for selecting from a list

Complete Student Registration Form:

```
<html>
<body>
<h2>Student Registration</h2>
<form action="register.jsp" method="POST">
Name: <input type="text" name="name"><br>
Email: <input type="text" name="email"><br>
Password: <input type="password" name="pwd"><br>
Gender:
<input type="radio" name="g" value="M"> Male
<input type="radio" name="g" value="F"> Female<br>
Hobbies:
<input type="checkbox" value="read"> Reading
<input type="checkbox" value="code"> Coding<br>
City:
<select name="city">
<option>Patna</option>
<option>Delhi</option>
</select><br>
<input type="submit" value="Register">
<input type="reset" value="Clear">
</form>
</body>
</html>
```

Conclusion:

HTML forms are essential for web interactivity. They enable user input for login pages, registration, search, and contact applications. Data is processed on the server using JSP, PHP, or other server-side languages.

SECTION A — MCQ (Multiple Choice Questions)**Q1. The top-level object in the JavaScript hierarchy is:****[PYQ: 2014]**

- ✓ A. Window object
- B. Document object
- C. Button object
- D. Form object

Explanation: window is the global/top-level object. All others (document, location, history) are its properties.

Q2. Which method joins two or more arrays into one?**[PYQ: 2014]**

- A. join()
- B. add()
- ✓ C. concat()
- D. merge()

Explanation: concat() merges arrays: [1,2].concat([3,4]) returns [1,2,3,4]. join() joins array to string.

Q3. Which method moves a window to a specific screen position?**[PYQ: 2014]**

- A. moveBy()
- ✓ B. moveTo()
- C. moveForm()
- D. position()

Explanation: moveTo(x,y) moves window to exact coordinates. moveBy() moves window relatively.

Q4. Which method removes focus from the current element?**[PYQ: 2014]**

- ✓ A. blur()
- B. alert()
- C. focus()
- D. defocus()

Explanation: blur() removes focus from an element. focus() sets focus. These are opposites.

Q5. Default client-side scripting language in HTML is:**[PYQ: 2018]**

- A. Java
- B. Perl
- C. EcmaScript
- ✓ D. JavaScript

Explanation: JavaScript is the default and most widely used client-side scripting language for HTML pages.

Q6. JavaScript code is typically triggered by:

[PYQ: 2019]

- A. RMI call
- ✓ **B. A browser event**
- C. A preprocessor
- D. Compiler

Explanation: JavaScript is event-driven. It runs when events like onclick, onsubmit, onchange occur.

Q7. Correct JavaScript syntax to write Hello World on web page:

[PYQ: 2020]

- A. System.out.println(...)
- B. println(...)
- ✓ **C. document.write("Hello")**
- D. response.write(...)

Explanation: document.write() outputs text/HTML directly onto the web page.

Q8. The 'this' keyword in JavaScript refers to:

[PYQ: 2020]

- ✓ **A. Current object**
- B. Previous object
- C. A variable
- D. Global object

Explanation: 'this' refers to the object that is calling/owning the current function or method.

Q9. JavaScript code is placed inside which HTML element?

[PYQ: 2020]

- A. <js>
- B. <scripting>
- ✓ **C. <script>**
- D. <javascript>

Explanation: JavaScript goes inside <script> tags. Can be in <head> or <body> section.

Q10. Main advantage of JavaScript in web development:

[PYQ: 2024, 2025]

- A. Server-side database access
- ✓ **B. Client-side interactivity**
- C. OS-level programming
- D. Hardware control

Explanation: JS provides client-side interactivity and real-time response without server trips.

SECTION B — SHORT ANSWER QUESTIONS (4 Marks)

Q1. Write a JavaScript program to check whether a number is Prime or not.**[PYQ: 2014]**

A Prime Number is divisible only by 1 and itself. Examples: 2, 3, 5, 7, 11, 13.

```
<html>
<body>
<script>
var n = parseInt(prompt("Enter a number:"));
var isPrime = true;
if(n <= 1) isPrime = false;
for(var i = 2; i <= Math.sqrt(n); i++){
  if(n % i == 0){
    isPrime = false;
    break;
  }
}
if(isPrime)
  document.write(n + " is a PRIME number");
else
  document.write(n + " is NOT a prime number");
</script>
</body>
</html>
```

Q2. Write a JavaScript program to print Fibonacci series up to N terms.**[PYQ: 2018]**

Fibonacci Series: Each number is sum of previous two. Series: 0, 1, 1, 2, 3, 5, 8, 13...

```
<html>
<body>
<script>
var n = parseInt(prompt("How many terms?"));
var a = 0, b = 1, c;
document.write("Fibonacci: " + a + ", " + b);
for(var i = 2; i < n; i++){
  c = a + b;
  document.write(", " + c);
  a = b;
  b = c;
}
</script>
</body>
</html>
```

Q3. Write a JavaScript program to check if a number is Odd or Even.**[PYQ: 2019]**

A number is Even if divisible by 2 (remainder = 0). Otherwise it is Odd.

```
<html>
<body>
<script>
var n = parseInt(prompt("Enter a number:"));
if(n % 2 == 0)
document.write(n + " is an EVEN number");
else
document.write(n + " is an ODD number");
</script>
</body>
</html>
```

Q4. Write a JavaScript program to find Factorial of a number.**[PYQ: 2022]**

Factorial of $n = n \times (n-1) \times \dots \times 2 \times 1$. Example: Factorial of 5 = $5 \times 4 \times 3 \times 2 \times 1 = 120$.

```
<html>
<body>
<script>
var n = parseInt(prompt("Enter a number:"));
var fact = 1;
if(n < 0)
document.write("Factorial not defined for negative numbers");
else{
for(var i = 1; i <= n; i++){
fact = fact * i;
}
document.write("Factorial of " + n + " = " + fact);
}
</script>
</body>
</html>
```

Q5. What are advantages of JavaScript? How to insert JavaScript in HTML? (★ 6 times)

[PYQ: 2017, 2018]

Advantages of JavaScript:

- **Client-side execution:** runs in browser — reduces load on server
- **Fast response:** no round-trip to server for basic operations
- **Interactive:** creates animations, popups, dynamic content
- **Form Validation:** validates user input before sending to server
- **Platform independent:** works on all browsers and operating systems
- **Easy to learn:** simple C-like syntax, no compilation needed

3 Ways to Insert JavaScript in HTML:

- **Internal:** code inside <script> tags within the HTML file
- **External:** code in separate .js file, linked with <script src="file.js"></script>
- **Inline:** code directly in HTML event attributes like onclick="function()"

Q6. How to declare and access arrays in JavaScript? Explain with array methods.

[PYQ: 2019]

An Array stores multiple values in a single variable. Array index starts from 0.

Declaring an Array:

```
var marks = [85, 90, 75, 88, 92];
```

Accessing Elements:

```
document.write(marks[0]); // Outputs: 85 (first element)
document.write(marks[4]); // Outputs: 92 (last element)
```

Important Array Methods:

- **concat():** joins two arrays — arr1.concat(arr2)
- **sort():** sorts array elements in order
- **push():** adds element at the end of array
- **pop():** removes last element from array
- **join():** joins all elements into one string
- **length:** returns total number of elements in array

SECTION C — LONG ANSWER QUESTIONS (12 Marks)

Introduction:

A JavaScript calculator performs four basic arithmetic operations: Addition, Subtraction, Multiplication, and Division.

```
<html>
<body>
<h2>JavaScript Calculator</h2>
<script>
function calculate(){
var n1 = parseFloat(document.getElementById("n1").value);
var n2 = parseFloat(document.getElementById("n2").value);
var op = document.getElementById("op").value;
var result;
if(op == "+") result = n1 + n2;
else if(op == "-") result = n1 - n2;
else if(op == "*") result = n1 * n2;
else if(op == "/"){
if(n2 == 0) result = "Error: Cannot divide by zero!";
else result = n1 / n2;
}
document.getElementById("res").innerHTML = "Result = " + result;
}
</script>
Number 1: <input type="text" id="n1"><br>
Number 2: <input type="text" id="n2"><br>
Operator:
<select id="op">
<option value="+">+ Addition</option>
<option value="-">- Subtraction</option>
<option value="*">* Multiplication</option>
<option value="/">/ Division</option>
</select><br>
<button onclick="calculate()">Calculate</button>
<p id="res"></p>
</body>
</html>
```

Conclusion:

This calculator uses getElementById() to get values, parseFloat() to convert strings to numbers, and innerHTML to display the result on page.

Q2. What is JavaScript? Explain advantages, disadvantages, data types, and operators. (★ 6 times) [PYQ: 2017, 2020, 2022, 2024, 2025]

Introduction:

JavaScript is a lightweight, interpreted, client-side scripting language developed by Brendan Eich at Netscape in 1995. It makes web pages dynamic and interactive.

Advantages of JavaScript:

- **Client-side:** runs in browser, reduces server load
- **Speed:** fast response without server requests
- **Interactivity:** animations, real-time validation, popups
- **Form Validation:** validates input before sending to server
- **Easy to learn:** no compilation, simple C-like syntax
- **Platform independent:** works on all browsers and OS

Disadvantages of JavaScript:

- **Security:** client-side code is visible to everyone (source code)
- **No database access:** cannot directly connect to a database
- **Browser compatibility:** minor differences between browsers
- **No file read/write:** cannot access files on user machine directly

Data Types in JavaScript:

- **String:** text data — var name = "AKU Patna";
- **Number:** numeric data — var age = 20;
- **Boolean:** true or false — var pass = true;
- **Null:** intentionally empty — var x = null;
- **Undefined:** variable declared but no value assigned — var y;
- **Array:** multiple values — var arr = [1, 2, 3];

Operators in JavaScript:

- **Arithmetic:** +, - (subtract), * (multiply), / (divide), % (modulus), ++ (increment), -- (decrement)
- **Comparison:** == (equal), != (not equal), > (greater), < (less), >= (greater or equal), === (strictly equal)
- **Logical:** && (AND), || (OR), ! (NOT)
- **Assignment:** =, +=, -=, *=, /=
- **Ternary:** condition ? value1 : value2

Conclusion:

JavaScript is the most popular programming language for web development. It powers interactive websites and is essential knowledge for every web developer.

Q3. Explain JavaScript standard library functions. Write JavaScript for form validation (Name, Mobile, Age).

[PYQ: 2019]

Standard Library Functions:

- **parseInt(str):** converts string to integer — parseInt("42") gives 42
- **parseFloat(str):** converts string to decimal — parseFloat("3.14") gives 3.14
- **isNaN(x):** returns true if x is Not a Number — isNaN("abc") gives true
- **Math.sqrt(n):** returns square root — Math.sqrt(16) gives 4
- **Math.abs(n):** returns absolute (positive) value
- **String.length:** gives number of characters in string

Form Validation Program:

```
<html>
<body>
<script>
function validate(){
var name = document.getElementById("n").value;
var mobile = document.getElementById("m").value;
var age = document.getElementById("a").value;
if(name == ""){
alert("Name cannot be empty!"); return false;
}
if(mobile.length != 10 || isNaN(mobile)){
alert("Mobile must be 10 digits!"); return false;
}
if(isNaN(age) || age < 0){
alert("Please enter valid age!"); return false;
}
alert("Form submitted successfully!");
return true;
}
</script>
Name: <input type="text" id="n"><br>
Mobile: <input type="text" id="m"><br>
Age: <input type="text" id="a"><br>
<button onclick="validate()">Submit</button>
</body>
</html>
```

SECTION A — MCQ (Multiple Choice Questions)

Q1. Which XML parser provides a callback (event-based) interface?**[PYQ: 2014]**

- A. DOM
- ✓ B. SAX
- C. Both DOM and SAX
- D. Neither

Explanation: SAX (Simple API for XML) = event-based callback parsing. DOM loads entire tree in memory.

Q2. An XML document must have:**[PYQ: 2020]**

- ✓ A. One root element
- B. One child element only
- C. All lowercase tags
- D. Attributes in double quotes only

Explanation: XML rule: exactly ONE root element must wrap all other elements in the document.

Q3. A dynamic web page is:**[PYQ: 2019]**

- A. Same every time displayed
- ✓ B. Generated on demand by server/program
- C. Shows only images
- D. Has no HTML

Explanation: Dynamic pages are generated at request time using server technologies like JSP, PHP, ASP.

Q4. SGML is the parent language of:**[PYQ: Syllabus]**

- A. HTML only
- B. XML only
- ✓ C. Both HTML and XML
- D. None

Explanation: SGML (1986) is the parent meta-language. Both HTML and XML are derived from/simplified subsets of SGML.

SECTION B — SHORT ANSWER QUESTIONS (4 Marks)

Q1. What is XML Schema (XSD)? Explain its purpose.**[PYQ: 2014]**

An XML Schema (XSD = XML Schema Definition) defines the structure, allowed elements, data types, and rules for a valid XML document. It acts as a blueprint for XML documents.

Purpose of XML Schema:

- Defines which elements and attributes are allowed in XML
- Specifies data types for elements (string, integer, date)
- Validates XML documents — checks if they follow the rules
- Makes data exchange between different systems reliable

Key Difference: DTD vs XSD:

- **DTD (Document Type Definition):** older, simpler method to validate XML
- **XSD:** modern, more powerful, supports data types and namespaces

Q2. What is the difference between XML and HTML?**[PYQ: 2018]**

Feature	XML	HTML
Full Form	eXtensible Markup Language	HyperText Markup Language
Purpose	Store and transport data	Display data in browser
Tags	User-defined (custom tags)	Predefined standard tags
Case Sensitive	YES — Name and name are different	NO — not case-sensitive
Closing Tags	Mandatory for all tags	Some optional (e.g.)
Root Element	Must have exactly one root	Has <html> as root

Q3. What is SGML? Explain difference between SGML and XML.**[PYQ: 2019, 2024]**

SGML (Standard Generalized Markup Language) is a meta-language used to define other markup languages. It was standardized by ISO in 1986. Both HTML and XML are derived from SGML.

Features of SGML:

- Very powerful and complex language
- Used for large document management systems (aerospace, publishing)
- Too complex for web browsers to implement fully
- Allows user-defined markup languages

SGML vs XML Differences:

- **Complexity:** SGML is very complex; XML is simplified subset of SGML
- **Usage:** SGML for large document systems; XML for web and data exchange
- **Tag rules:** SGML allows tag omission; XML does NOT allow omission
- **Well-formedness:** XML must be well-formed with strict rules
- **Ease:** XML much easier to learn and use than SGML

SECTION C — LONG ANSWER QUESTIONS (12 Marks)

Q1. Explain SGML, XML and their differences. Write XML document for 5 books in a bookstore. (★ 4 times)

[PYQ: 2018, 2019, 2024, 2025]

Introduction:

XML (eXtensible Markup Language) is a flexible, self-describing markup language for storing and transporting data. Unlike HTML, XML tags are user-defined — you choose your own tag names.

Rules for Well-Formed XML:

- Must have exactly ONE root element wrapping all others
- **All tags must be properly closed:** <name>...</name>
- **Tags are CASE-SENSITIVE:** <Name> is different from <name>
- **Attribute values must be in double quotes:** id="1"
- Tags must be properly nested — no overlapping of tags
- XML declaration must be on the very first line

XML Document — Bookstore with 5 Books:

```
<?xml version="1.0" encoding="UTF-8"?>
<bookstore>
  <book id="1">
    <title>Java Programming</title>
    <author>James Gosling</author>
    <price>450</price>
  </book>
  <book id="2">
    <title>HTML and CSS Design</title>
    <author>Jon Duckett</author>
    <price>380</price>
  </book>
  <book id="3">
    <title>JavaScript: The Good Parts</title>
    <author>Douglas Crockford</author>
    <price>320</price>
  </book>
  <book id="4">
    <title>Introduction to Algorithms</title>
    <author>Cormen et al.</author>
    <price>650</price>
  </book>
  <book id="5">
    <title>Web Designing with JSP</title>
    <author>Ivan Bay Ross</author>
    <price>290</price>
  </book>
</bookstore>
```

SGML vs XML vs HTML Comparison:

- **SGML (1986):** Parent meta-language; complex; for large document systems
- **HTML (1991):** Derived from SGML; for displaying data in web browsers
- **XML (1998):** Derived from SGML; simplified; for storing and transporting data

XML Parsers — DOM vs SAX:

- **DOM Parser:** loads entire XML document into memory as a tree structure; allows random access to any element
- **SAX Parser:** event-based; reads document sequentially; very memory efficient; provides callback interface

Conclusion:

XML is a platform-independent, self-describing language ideal for data exchange between different systems and platforms. It is widely used in web services, configuration files, and data storage.

SECTION A — MCQ (Multiple Choice Questions)

Q1. JSP stands for:

[PYQ: 2025]

- A. JavaScript Pages
- B. JavaServer Protocols
- ✓ C. JavaServer Pages
- D. Java System Programs

Explanation: JSP = JavaServer Pages — a server-side Java technology to create dynamic web pages.

Q2. Which of the following is a server-side technology?

[PYQ: 2018]

- A. HTML
- ✓ B. JSP
- C. JavaScript
- D. CSS

Explanation: JSP runs on the server. HTML, CSS, JavaScript all run on the client (browser) side.

Q3. Which is NOT a valid attribute of the JSP page directive?

[PYQ: 2019]

- A. language
- B. extends
- ✓ C. export
- D. import

Explanation: Export is NOT a valid page directive attribute. Valid ones: language, import, session, errorPage, extends.

Q4. Which JSP element directly outputs the value of an expression?

[PYQ: 2025]

- A. <%! ... %>
- B. <% ... %>
- ✓ C. <%= ... %>
- D. <%@ ... %>

Explanation: <%= expression %> is the JSP expression element — evaluates and outputs the value directly.

SECTION B — SHORT ANSWER QUESTIONS (4 Marks)

Q1. What are JSP Standard Actions? Explain each with syntax. (★ 6 times)**[PYQ: 2017, 2020]**

JSP Standard Actions are XML-based tags that control the JSP engine behavior at request time (runtime). They perform tasks like including files, forwarding requests, and working with JavaBeans.

All Standard Actions:

- `<jsp: include page="file.jsp">`: Dynamically includes another JSP page output at request time
- `<jsp: forward page="page.jsp">`: Forwards the request and response to another resource
- `<jsp: param name="x" value="y">`: Passes parameter values with include or forward actions
- `<jsp: useBean id="b" class="MyBean" scope="session">`: Creates or locates a JavaBean instance
- `<jsp: setProperty name="b" property="name" value="Rahul">`: Sets a property value of a JavaBean
- `<jsp: getProperty name="b" property="name">`: Gets and outputs a property value from a JavaBean
- `<jsp: plugin>`: Embeds a Java Applet or Bean in the browser using object or embed element

Q2. What are JSP scripting elements? Explain all four types with examples.**[PYQ: 2019, 2020]**

JSP Scripting Elements allow embedding Java code directly inside JSP pages. There are four types:

1. Scriptlet — `<% ... %>`:

- Contains Java statements executed on each client request
- **Example** — `<% int x = 10; out.println("Value: " + x); %>`

2. Expression — `<%= ... %>`:

- Evaluates a Java expression and outputs its value (no semicolon inside)
- **Example** — Hello, `<%= request.getParameter("name") %>`

3. Declaration — `<%! ... %>`:

- Declares variables or methods at class level (outside `_jspService()` method)
- **Example** — `<%! int count = 0; String greet(String n){ return "Hi "+n; } %>`

4. Comment — `<%-- ... --%>`:

- JSP comment — NOT sent to browser (unlike HTML comment `<!-- -->`)
- **Example** — `<%-- This is a server-side comment --%>`

Q3. What are the life-cycle methods of JSP? Explain each step.

[PYQ: 2020]

The JSP Life Cycle is managed by the web container (Apache Tomcat). It has 5 main stages from creation to destruction:

JSP Life Cycle — Step by Step:

- **Step 1 — Translation:** JSP file (.jsp) is converted to a Java Servlet file (.java) by the container
- **Step 2 — Compilation:** The generated .java file is compiled into a .class file (bytecode)
- **Step 3 — `jspInit()`:** Called ONCE when JSP is first loaded. Used for one-time initialization (like DB connection)
- **Step 4 — `_jspService(request, response)`:** Called for EVERY client request. Contains the main page logic.
- **Step 5 — `jspDestroy()`:** Called ONCE when JSP is removed from service. Used for cleanup (close connections).

Life Cycle Order:

- Translate then Compile then `jspInit()` then `_jspService()` (for each request) then `jspDestroy()`

Q4. What is Page Directive? What is Include Directive? Explain with examples.

[PYQ: 2018, 2022]

JSP Directives give instructions to the JSP container at translation time (when JSP is first converted to Java). They use the syntax: `<%@ directive attribute="value" %>`

1. Page Directive — `<%@ page ... %>`:

- Applies settings and attributes to the entire JSP page
- **language:** scripting language used (default: java)
- **import:** Java classes to import (like import in Java files)
- **session:** true or false — enables/disables session tracking
- **errorPage:** URL of page to show if an error occurs
- **contentType:** MIME type and character set (text/html; charset=UTF-8)

2. Include Directive — `<%@ include file="..." %>`:

- Statically includes content of another file at translation time
- The included file becomes part of the JSP before compilation
- Example — `<%@ include file="header.html" %>` at top, `<%@ include file="footer.html" %>` at bottom
- **Difference from `jsp:include`:** include: directive is static (compile time); `jsp:include` is dynamic (runtime)

SECTION C — LONG ANSWER QUESTIONS (12 Marks)

Q1. Explain all JSP Standard Actions in detail with syntax and examples. (★ [PYQ: 2014, 2017, 2018, 2019, 2020, 2022])
MOST REPEATED — 6 times)

Introduction:

JSP Standard Actions are special XML-based tags built into JSP that control the JSP engine behavior at runtime. They provide ready-made functionality for common tasks without writing Java code directly.

1. jsp:include — Dynamic Include:

- Includes the output of another page at request time (dynamic)
- Can pass parameters with jsp: param sub-element

```
<jsp:include page="header.jsp">
<jsp:param name="user" value="Rahul"/>
</jsp:include>
```

2. jsp:forward — Forward Request:

- Forwards the entire request to another JSP, HTML page, or servlet
- Current page stops processing and target page takes over

```
<jsp:forward page="welcome.jsp">
<jsp:param name="msg" value="Login OK"/>
</jsp:forward>
```

3. jsp:useBean — Create or Find JavaBean:

- Creates or locates a JavaBean instance with given scope
- **scope:** page, request, session, application

```
<jsp:useBean id="student" class="com.StudentBean" scope="session"/>
```

4. jsp:setProperty — Set Bean Property:

- Sets a value for one or all properties of a JavaBean

```
<jsp:setProperty name="student" property="name" value="Rahul Kumar"/>
```

5. jsp:getProperty — Get Bean Property:

- Gets a property value from a JavaBean and outputs it to page

```
Student Name: <jsp:getProperty name="student" property="name"/>
```

6. jsp:param — Pass Parameters:

- Used inside include or forward to pass additional parameters

7. jsp:plugin — Java Plugin:

- Embeds a Java Applet in the browser using HTML object or embed element

Conclusion:

JSP Standard Actions provide powerful runtime functionality through simple XML-style tags. They make JSP pages more maintainable and cleaner than embedding complex Java scriptlet code directly.

Q2. What is JSP? Explain advantages. Write HTML form and JSP page for student registration.

[PYQ: 2017, 2018, 2020, 2022, 2025]

Introduction:

JSP (JavaServer Pages) is a server-side technology that allows embedding Java code within HTML pages to generate dynamic web content. JSP pages run on the server and are compiled into Servlets by the web container (Apache Tomcat).

Advantages of JSP:

- **Dynamic Content:** generates different HTML based on user input or database data
- **Full Java Power:** access to all Java APIs, JDBC for database, JavaBeans
- **Platform Independent:** runs on any server with JVM (Write Once, Run Anywhere)
- **MVC Support:** JSP for View, Servlet for Controller — clean separation
- **Session Management:** built-in session, request, response objects available
- **Database Access:** can use JDBC to connect and query databases
- **Reusable Components:** JavaBeans and custom Tag Libraries

HTML Form — form.html:

```
<html>
<body bgcolor="lightblue">
<h2>Student Registration Form</h2>
<form action="register.jsp" method="POST">
<table border="1">
<tr><td>First Name:</td>
<td><input type="text" name="fname"></td></tr>
<tr><td>Last Name:</td>
<td><input type="text" name="lname"></td></tr>
<tr><td>Password:</td>
<td><input type="password" name="pwd"></td></tr>
<tr><td colspan="2">
<input type="submit" value="Register">
</td></tr>
</table>
</form>
</body>
</html>
```

JSP Processing Page — register.jsp:

```
<%@ page language="java" contentType="text/html" %>

<html>

<body bgcolor="lightyellow">

<h2>Registration Successful!</h2>

<%

String fname = request.getParameter("fname");
String lname = request.getParameter("lname");

%>

<table border="1" cellpadding="8">
<tr><th>Field</th><th>Value</th></tr>
<tr><td>First Name</td><td><%= fname %></td></tr>
<tr><td>Last Name</td><td><%= lname %></td></tr>
</table>

</body>

</html>
```

Conclusion:

JSP combines Java power with HTML simplicity to create dynamic, data-driven web applications. It is widely used for building enterprise-level web applications on the Java EE platform.

MOST REPEATED QUESTIONS — PYQ 2014 to 2025

#	Topic / Question	Unit	Times	PYQ Years
1	TCP/IP Layers and Services (Full explanation) ★VVI	Unit 1	7x	2017,2018,2019,2020,2022,2024,2025
2	JSP Standard Actions (jsp:include, forward, useBean) ★VVI	Unit 5	6x	2014,2017,2018,2019,2020,2022
3	JavaScript Advantages and Disadvantages ★VVI	Unit 3	6x	2017,2018,2020,2022,2024,2025
4	HTML Structure (What is HTML, document structure) ★VVI	Unit 2	5x	2018,2019,2020,2024,2025
5	Page Directive in JSP (all attributes) ★VVI	Unit 5	5x	2014,2018,2019,2022,2025
6	JS Programs (Prime, Factorial, Fibonacci, Odd/Even) ★VVI	Unit 3	5x	2014,2018,2019,2020,2022
7	SGML vs XML, XML document with bookstore example	Unit 4	4x	2018,2019,2022,2024,2025
8	HTML Tables (all tags, colspan, rowspan)	Unit 2	4x	2018,2019,2020,2022
9	HTML Form elements with registration example	Unit 2	4x	2018,2019,2020,2022
10	JSP Scripting Elements (all four types)	Unit 5	4x	2019,2020,2022,2025
11	JSP Advantages + Registration Form (HTML + JSP)	Unit 5	4x	2017,2020,2022,2025
12	JavaScript Calculator program (all operations)	Unit 3	3x	2014,2019,2022
13	Domain Registration and Email Working	Unit 1	3x	2022,2024,2025
14	JavaScript Operators and Data Types	Unit 3	3x	2020,2022,2024

LAST MINUTE CHEAT SHEET

IMPORTANT FULL FORMS

- HTML = HyperText Markup Language
- HTTP = HyperText Transfer Protocol
- FTP = File Transfer Protocol
- TCP = Transmission Control Protocol
- IP = Internet Protocol
- URL = Uniform Resource Locator (NOT Universal!)
- JSP = JavaServer Pages
- XML = eXtensible Markup Language
- SGML = Standard Generalized Markup Language
- ISP = Internet Service Provider

- DNS = Domain Name System
- WWW = World Wide Web
- DOM = Document Object Model
- SAX = Simple API for XML

HTML TAGS QUICK REFERENCE

- <html> — Root element of the document
- <head> — Meta info (NOT visible on page)
- <body> — All visible page content goes here
- <h1> to <h6> — Heading tags (h1 = largest)
- <p> — Paragraph;
 — Line break
- <i> <u> — Bold, Italic, Underline
- — Image (NOT <image>!)
- — Hyperlink (anchor tag)
- <table><tr><th><td> — Tables
- — Unordered/Ordered Lists
- <form><input><select> — Forms
- <hr> — Horizontal rule/line
- <frameset><frame> — Frames
- <script> — JavaScript placement tag

TCP/IP LAYERS — TOP TO BOTTOM

- Layer 4: Application Layer
- Protocols: HTTP, FTP, SMTP, DNS, Telnet
- Layer 3: Transport Layer
- Protocols: TCP (reliable), UDP (fast)
- Layer 2: Internet / Network Layer
- Protocols: IP, ICMP, ARP
- Layer 1: Network Access Layer
- Protocols: Ethernet, Wi-Fi, PPP

JSP SCRIPTING ELEMENTS

- <% code %> — Scriptlet (Java code runs each request)
- <%= expr %> — Expression (outputs value, no semicolon)
- <%! decl %> — Declaration (class-level variable/method)
- <%@ page ... %> — Page directive (settings for page)
- <%@ include file="" %> — Include directive (static)
- <%-- comment --%> — JSP comment (not sent to browser)
- NOT valid: export, extend (page directive attributes)

JAVASCRIPT KEY POINTS

- Top-level object = window (global object)
- document.write() = outputs text on the page

- alert() = shows popup dialog box
- prompt() = shows input dialog box
- this = current object in the method
- NaN = Not a Number (use isNaN() to check)
- concat() joins two arrays together
- sort() sorts array elements
- blur() removes focus from element
- moveTo() moves browser window to position
- Default scripting language in HTML = JavaScript
- parseInt() converts string to integer

MEMORY TRICKS

- SGML is PARENT of both HTML and XML
- TCP = Reliable delivery; UDP = Fast delivery
- DOM = Tree structure in memory (random access)
- SAX = Event-based callbacks (sequential)
- HTML = for Display; XML = for Data storage
- Client-side = JavaScript; Server-side = JSP
- URL = Uniform (NOT Universal!) Resource Locator
- IPv4 = 32 bits; IPv6 = 128 bits
- blur() removes focus (blur makes things less clear!)
- Export is NOT valid page directive (remember: export!)
- JSP lifecycle: Translate-Compile-Init-Service-Destroy

COMMON EXAM MISTAKES

- WRONG: <image> tag | CORRECT: tag
- WRONG: Universal Resource Locator | CORRECT: Uniform
- WRONG: IPv6 = 32 bits | CORRECT: IPv6 = 128 bits
- WRONG: join() merges arrays | CORRECT: concat()
- WRONG: JSP is client-side | CORRECT: JSP is server-side
- WRONG: HTML is case-sensitive | CORRECT: XML is case-sensitive
- WRONG: Export is valid page directive | CORRECT: NOT valid
- WRONG: Dennis Ritchie invented WWW | CORRECT: Tim Berners-Lee
- WRONG: SAX loads entire document | CORRECT: DOM loads tree

QUICK EXAM ANSWERS

- Father of WWW = Tim Berners-Lee (1989, CERN)
- Web protocol = HTTP (HyperText Transfer Protocol)
- IPv4 = 32 bits | IPv6 = 128 bits
- Internet = Packet switching technology
- JSP = JavaServer Pages (server-side)
- Default scripting = JavaScript
- Window = top-level JavaScript object

- "this" = current object in JavaScript
- blur() = removes focus from element
- concat() = joins two or more arrays
- DOM = loads XML as tree, random access
- SAX = event-based callbacks, sequential